

Shagun Shaw

Backend Web Developer

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Summary

Backend-focused software engineering enthusiast with a strong foundation in server-side development and problem solving. Experienced in designing scalable APIs, managing databases, and building efficient backend architectures for reliable web applications.

Education

Techno India University

Bachelor of Computer Science and Engineering Aug, 2023 - Sept, 2027 GPA: 9.07

Rajasthan Vidya Mandir

Higher Secondary Apr, 2021 - Mar, 2023 Marks: 92.7%

Rajasthan Vidya Mandir

Secondary Apr, 2020 - Mar, 2021 Marks: 91.2%

Technical Skills

Python, Java, JavaScript, C++, Express, Node.js, AWS S3, Docker, HTML, CSS, NextJS, MySQL, MongoDB, Git, Postman, LLM-based App Development, RAG

Soft Skills

Collaboration, Problem-solving, Communication, Time management, Result-oriented

Languages

English, Hindi

Projects:

Project 1: QuizBlitz — Real Time Quiz Hosting Platform: TypeScript, Node.js, Express.js, Socket.io, MongoDB, JWT, Nodemailer, Winston

URL: <https://github.com/ShagunShaw/quizblitz>

- Engineered Google OAuth 2.0 from scratch without any auth library, implementing a stateful JWT system with refresh token rotation, multi-device session management via per-device session IDs, and secure httpOnly cookie handling
- Built a Socket.io powered real time quiz engine with isolated in-memory room state management, synchronized question broadcasting, and a server side timer system with a 2 second buffer to ensure all client submissions are collected before result processing
- Designed a custom scoring algorithm with order based point distribution (15pts to 3pts based on submission order), streak bonus multipliers (3/5/10 consecutive correct answers), and auto-submit handling for time expired responses
- Implemented a co-host invite system using JWT signed expiring email links via Nodemailer, handling separate flows for existing and new users with pending invite persistence in MongoDB
- Integrated Winston logger with Axiom for production monitoring and node-cron for automated quiz cleanup post-attempt

Project 2: SecureDrop – Ephemeral File Sharing Platform | NestJS, TypeScript, MongoDB, Cloudinary, Vanilla JS, HTML, CSS

URL: <https://github.com/ShagunShaw/Secure-Drop>

- Developed a full-stack, "burn-after-reading" file-sharing application featuring a static, drag-and-drop frontend and a modular NestJS backend to facilitate secure, temporary data exchange via unique access codes.
- Engineered a backend processing pipeline utilizing Multer memory storage and Archiver to automatically aggregate and compress multi-file uploads into a single unified ZIP bundle.
- Implemented a scheduled cron job service (@nestjs/schedule) that executes nightly at midnight to purge expired binary bundles from Cloudinary and delete corresponding metadata tracking records from MongoDB.
- Enforced rigid security measures by building custom validation pipes to block restricted file formats (.exe, .bat, .sh), imposing a 50 MB payload ceiling, and embedding tiered client-side rate limiting using @nestjs/throttler.

Achievements:

- Honoured with Medha Samman (Taaza TV) and Pratibha Samman (Prabhat Khabar) in 2023 for being the school topper in ISC 12th board examination.
- Participated in Smart India Hackathon 2024 and got selected in top 30 teams in the internal college round for innovative tech solution.
- Selected as a Core Team Member for GDG On Campus TIU in the DSA domain, contributing to technical activities, student sessions and community initiatives.